

Groups and Lie Algebras: Solutions to Exercise Sheet 7

1 Derivations of an associative algebra

Problem 1. Let A be a finite-dimensional associative algebra. A linear map $D \in \text{End}(A)$ is called a derivation of A if

$$D(xy) = D(x)y + xD(y)$$

for all $x, y \in A$. The vector space $\text{End}(A)$ has a Lie algebra structure $\mathfrak{gl}(A)$, where the bracket is the commutator

$$[S, T] = S \circ T - T \circ S.$$

Show that $\text{Der}(A)$, the space of derivations, is a Lie subalgebra of $\mathfrak{gl}(A)$. What is the corresponding Lie group whose Lie algebra is $\text{Der}(A)$?

Solution. We have to prove two things.

First, $\text{Der}(A)$ is a vector subspace of $\text{End}(A)$. Second, it is closed under the commutator bracket. Together these two statements say precisely that $\text{Der}(A)$ is a Lie subalgebra of $\mathfrak{gl}(A)$.

Step 1: $\text{Der}(A)$ is a vector subspace

Let $D_1, D_2 \in \text{Der}(A)$, and let $a, b \in \mathbb{C}$ or $a, b \in \mathbb{R}$, depending on the base field of A . Define

$$D = aD_1 + bD_2.$$

Then for $x, y \in A$,

$$\begin{aligned} D(xy) &= aD_1(xy) + bD_2(xy) \\ &= a(D_1(x)y + xD_1(y)) + b(D_2(x)y + xD_2(y)) \\ &= (aD_1(x) + bD_2(x))y + x(aD_1(y) + bD_2(y)) \\ &= D(x)y + xD(y). \end{aligned}$$

Thus D is again a derivation. Therefore $\text{Der}(A)$ is a vector subspace of $\text{End}(A)$.

Step 2: the commutator of two derivations is a derivation

Let $D_1, D_2 \in \text{Der}(A)$. We need to show that

$$[D_1, D_2] = D_1D_2 - D_2D_1$$

is again a derivation. That is, we must prove

$$[D_1, D_2](xy) = [D_1, D_2](x)y + x[D_1, D_2](y).$$

We compute carefully. First,

$$\begin{aligned} D_1 D_2(xy) &= D_1(D_2(x)y + xD_2(y)) \\ &= D_1(D_2(x)y) + D_1(xD_2(y)) \\ &= D_1 D_2(x)y + D_2(x)D_1(y) + D_1(x)D_2(y) + xD_1 D_2(y). \end{aligned}$$

Similarly,

$$\begin{aligned} D_2 D_1(xy) &= D_2(D_1(x)y + xD_1(y)) \\ &= D_2 D_1(x)y + D_1(x)D_2(y) + D_2(x)D_1(y) + xD_2 D_1(y). \end{aligned}$$

Subtracting the two expressions, the mixed terms cancel:

$$\begin{aligned} [D_1, D_2](xy) &= (D_1 D_2(x) - D_2 D_1(x))y + x(D_1 D_2(y) - D_2 D_1(y)) \\ &= [D_1, D_2](x)y + x[D_1, D_2](y). \end{aligned}$$

Therefore $[D_1, D_2] \in \text{Der}(A)$. Hence $\text{Der}(A)$ is closed under the commutator bracket.

We conclude that

$$\text{Der}(A) \subset \mathfrak{gl}(A)$$

is a Lie subalgebra.

The corresponding Lie group

The natural Lie group whose Lie algebra is $\text{Der}(A)$ is the automorphism group of the associative algebra A :

$$\text{Aut}(A) = \{\Phi \in \text{GL}(A) \mid \Phi(xy) = \Phi(x)\Phi(y) \text{ for all } x, y \in A\}.$$

Here $\text{GL}(A)$ denotes the group of invertible linear maps $A \rightarrow A$. Thus $\text{Aut}(A)$ consists of the invertible linear maps which preserve the algebra multiplication.

Let us see why its Lie algebra is $\text{Der}(A)$. Suppose

$$\Phi(t) \in \text{Aut}(A), \quad \Phi(0) = \text{id}_A,$$

is a smooth curve of algebra automorphisms. Let

$$D = \Phi'(0).$$

Since each $\Phi(t)$ preserves multiplication, we have

$$\Phi(t)(xy) = \Phi(t)(x)\Phi(t)(y).$$

Differentiate at $t = 0$:

$$D(xy) = D(x)y + xD(y).$$

Thus every tangent vector at the identity of $\text{Aut}(A)$ is a derivation.

Conversely, derivations are precisely the infinitesimal algebra automorphisms. In finite dimensions, one can exponentiate a derivation by

$$\exp(tD) = \sum_{k=0}^{\infty} \frac{t^k D^k}{k!}.$$

The derivation property implies that $\exp(tD)$ preserves multiplication:

$$\exp(tD)(xy) = \exp(tD)(x)\exp(tD)(y).$$

So $\exp(tD) \in \text{Aut}(A)$, at least locally in the Lie group sense. Therefore

$$\text{Lie}(\text{Aut}(A)) = \text{Der}(A).$$

Thus the corresponding Lie group is

$$\boxed{\text{Aut}(A)}.$$

2 Representations of Lie algebras

Problem 2. Let $\mathfrak{g}$ be a Lie algebra and let V, W be representation spaces. Show that:

(a) there is a representation on $V \oplus W$ given by

$$((R_V \oplus R_W)(x))(v \oplus w) := R_V(x)v \oplus R_W(x)w;$$

(b) there is a representation on $V \otimes W$ given by

$$((R_V \otimes R_W)(x))(v \otimes w) := R_V(x)v \otimes w + v \otimes R_W(x)w;$$

(c) the following formula does not define a representation on $V \otimes W$:

$$x \mapsto (v \otimes w \mapsto R_V(x)v \otimes R_W(x)w);$$

(d) if V, W are induced by representations of a Lie group G with Lie algebra $\mathfrak{g}$, then the definitions above are compatible with the direct sum and tensor product of G -representations.

Solution. Recall that a representation of a Lie algebra $\mathfrak{g}$ on a vector space V is a Lie algebra homomorphism

$$R_V : \mathfrak{g} \longrightarrow \mathfrak{gl}(V).$$

This means two things:

$$R_V \text{ is linear,}$$

and

$$R_V([x, y]) = [R_V(x), R_V(y)]$$

for all $x, y \in \mathfrak{g}$. Similarly for R_W .

Part (a): direct sums

Define

$$R_{V \oplus W}(x) = R_V(x) \oplus R_W(x).$$

In block matrix form, relative to the decomposition $V \oplus W$, this is

$$R_{V \oplus W}(x) = \begin{pmatrix} R_V(x) & 0 \\ 0 & R_W(x) \end{pmatrix}.$$

Linearity in x is immediate because both R_V and R_W are linear.

Now we check the bracket condition. For $x, y \in \mathfrak{g}$,

$$\begin{aligned} [R_{V \oplus W}(x), R_{V \oplus W}(y)] &= \left[\begin{pmatrix} R_V(x) & 0 \\ 0 & R_W(x) \end{pmatrix}, \begin{pmatrix} R_V(y) & 0 \\ 0 & R_W(y) \end{pmatrix} \right] \\ &= \begin{pmatrix} [R_V(x), R_V(y)] & 0 \\ 0 & [R_W(x), R_W(y)] \end{pmatrix} \\ &= \begin{pmatrix} R_V([x, y]) & 0 \\ 0 & R_W([x, y]) \end{pmatrix} \\ &= R_{V \oplus W}([x, y]). \end{aligned}$$

Therefore $V \oplus W$ is a representation of $\mathfrak{g}$.

Part (b): tensor products

Define

$$R_{V \otimes W}(x) := R_V(x) \otimes \text{id}_W + \text{id}_V \otimes R_W(x).$$

On pure tensors this means

$$R_{V \otimes W}(x)(v \otimes w) = R_V(x)v \otimes w + v \otimes R_W(x)w.$$

Linearity in x is again immediate. The key point is the bracket condition. Let

$$A_x = R_V(x) \otimes \text{id}_W, \quad B_x = \text{id}_V \otimes R_W(x).$$

Then

$$R_{V \otimes W}(x) = A_x + B_x.$$

Since A_x acts only on the first tensor factor and B_y acts only on the second tensor factor, they commute:

$$[A_x, B_y] = 0$$

for all x, y . Therefore

$$\begin{aligned} [R_{V \otimes W}(x), R_{V \otimes W}(y)] &= [A_x + B_x, A_y + B_y] \\ &= [A_x, A_y] + [B_x, B_y]. \end{aligned}$$

But

$$[A_x, A_y] = [R_V(x), R_V(y)] \otimes \text{id}_W = R_V([x, y]) \otimes \text{id}_W,$$

and

$$[B_x, B_y] = \text{id}_V \otimes [R_W(x), R_W(y)] = \text{id}_V \otimes R_W([x, y]).$$

Hence

$$\begin{aligned} [R_{V \otimes W}(x), R_{V \otimes W}(y)] &= R_V([x, y]) \otimes \text{id}_W + \text{id}_V \otimes R_W([x, y]) \\ &= R_{V \otimes W}([x, y]). \end{aligned}$$

Thus this formula defines a representation on $V \otimes W$.

Part (c): why the naive formula is wrong

The proposed formula is

$$\tilde{R}(x)(v \otimes w) = R_V(x)v \otimes R_W(x)w.$$

The first problem is that this is generally not linear in x . A Lie algebra representation must be a linear map

$$\mathfrak{g} \rightarrow \mathfrak{gl}(V \otimes W).$$

Here is a very simple counterexample. Let $\mathfrak{g} = \mathbb{C}e$ be the one-dimensional abelian Lie algebra, and let $V = W = \mathbb{C}$ be the one-dimensional representation given by

$$R_V(e) = 1, \quad R_W(e) = 1.$$

Then for $x = ae$,

$$R_V(ae) = a, \quad R_W(ae) = a.$$

The proposed formula gives

$$\tilde{R}(ae) = a \cdot a = a^2$$

on $V \otimes W \cong \mathbb{C}$. But a linear map in $x = ae$ should scale by a , not by a^2 . For example,

$$\tilde{R}(2e) = 4 \neq 2 = 2\tilde{R}(e).$$

So the proposed construction is not even linear in x , and hence cannot be a Lie algebra representation.

Conceptually, this is why the correct formula contains a sum:

$$R_{V \otimes W}(x) = R_V(x) \otimes \text{id}_W + \text{id}_V \otimes R_W(x).$$

It is the infinitesimal version of the group-level tensor product action.

Part (d): compatibility with Lie group representations

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Suppose that $\Pi_V : G \rightarrow \text{GL}(V)$ and $\Pi_W : G \rightarrow \text{GL}(W)$ are representations of G . Their derived Lie algebra representations are

$$R_V(x) = \left. \frac{d}{dt} \right|_{t=0} \Pi_V(\exp(tx)),$$

and similarly for W .

For the direct sum, the group representation is

$$\Pi_{V \oplus W}(g) = \Pi_V(g) \oplus \Pi_W(g).$$

Differentiating at the identity gives

$$\begin{aligned} R_{V \oplus W}(x) &= \left. \frac{d}{dt} \right|_{t=0} (\Pi_V(\exp(tx)) \oplus \Pi_W(\exp(tx))) \\ &= R_V(x) \oplus R_W(x). \end{aligned}$$

This is exactly the formula from part (a).

For the tensor product, the group representation is

$$\Pi_{V \otimes W}(g) = \Pi_V(g) \otimes \Pi_W(g).$$

Thus

$$\Pi_{V \otimes W}(\exp(tx)) = \Pi_V(\exp(tx)) \otimes \Pi_W(\exp(tx)).$$

Differentiating at $t = 0$ and using the product rule gives

$$\begin{aligned} R_{V \otimes W}(x) &= \left. \frac{d}{dt} \right|_{t=0} (\Pi_V(\exp(tx)) \otimes \Pi_W(\exp(tx))) \\ &= R_V(x) \otimes \text{id}_W + \text{id}_V \otimes R_W(x). \end{aligned}$$

On pure tensors this is

$$R_{V \otimes W}(x)(v \otimes w) = R_V(x)v \otimes w + v \otimes R_W(x)w.$$

So the Lie algebra formulas are precisely the infinitesimal versions of the corresponding Lie group formulas.

3 Angular momentum operators in quantum mechanics

Problem 3. (a) Let $\mathfrak{g}$ be a real Lie algebra. Its complexification is

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{g} \oplus i\mathfrak{g},$$

with the $\mathbb{C}$ -bilinear extension of the Lie bracket. Compute the Lie algebra of the group $\mathrm{SL}(n)$ of complex matrices with determinant 1 and show that it is isomorphic to the complexification of $\mathfrak{su}(n)$.

(b) Consider $L^2(\mathbb{R}^3)$ with the usual position operators x_i and momentum operators $p_i = -i\frac{\partial}{\partial x_i}$. Show that the angular momentum operators

$$L_i = \sum_{j,k} \epsilon_{ijk} x_j p_k$$

form a representation of the complex Lie algebra $\mathfrak{sl}(2, \mathbb{C})$.

(c) Describe the decomposition into irreducible representations using the spherical harmonics from the previous exercise sheet.

Solution. Part (a): $\mathrm{Lie}(\mathrm{SL}(n, \mathbb{C}))$ and the complexification of $\mathfrak{su}(n)$

The group $\mathrm{SL}(n, \mathbb{C})$ is

$$\mathrm{SL}(n, \mathbb{C}) = \{A \in \mathrm{GL}(n, \mathbb{C}) \mid \det A = 1\}.$$

Its Lie algebra is the tangent space at the identity. Let $A(t)$ be a smooth curve in $\mathrm{SL}(n, \mathbb{C})$ such that

$$A(0) = I, \quad A'(0) = X.$$

Since $A(t) \in \mathrm{SL}(n, \mathbb{C})$, we have

$$\det A(t) = 1$$

for all t . Differentiating at $t = 0$ gives

$$\left. \frac{d}{dt} \right|_{t=0} \det A(t) = 0.$$

The derivative of the determinant at the identity is the trace:

$$(d \det)_I(X) = \mathrm{tr} X.$$

Therefore

$$\mathrm{tr} X = 0.$$

Thus

$$\mathrm{Lie}(\mathrm{SL}(n, \mathbb{C})) \subseteq \{X \in \mathrm{Mat}_n(\mathbb{C}) \mid \mathrm{tr} X = 0\}.$$

Conversely, suppose $X \in \mathrm{Mat}_n(\mathbb{C})$ satisfies $\mathrm{tr} X = 0$. Then

$$\det(e^{tX}) = e^{t \mathrm{tr} X} = 1.$$

So $e^{tX} \in \mathrm{SL}(n, \mathbb{C})$, and

$$\left. \frac{d}{dt} \right|_{t=0} e^{tX} = X.$$

Hence every traceless complex matrix occurs as a tangent vector at the identity. Therefore

$$\text{Lie}(\text{SL}(n, \mathbb{C})) = \mathfrak{sl}(n, \mathbb{C}) = \{X \in \text{Mat}_n(\mathbb{C}) \mid \text{tr } X = 0\}.$$

Now we compare this with $\mathfrak{su}(n)$. Recall that

$$\text{SU}(n) = \{A \in \text{U}(n) \mid \det A = 1\}.$$

Its Lie algebra is

$$\mathfrak{su}(n) = \{X \in \text{Mat}_n(\mathbb{C}) \mid X^\dagger + X = 0, \text{tr } X = 0\}.$$

Thus $\mathfrak{su}(n)$ consists of traceless skew-Hermitian matrices.

We claim that

$$\mathfrak{su}(n)_{\mathbb{C}} := \mathfrak{su}(n) \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{sl}(n, \mathbb{C}).$$

To see this concretely, we show that every traceless complex matrix can be written uniquely as

$$Z = X + iY, \quad X, Y \in \mathfrak{su}(n).$$

Let $Z \in \mathfrak{sl}(n, \mathbb{C})$. Define

$$X = \frac{Z - Z^\dagger}{2}, \quad Y = \frac{Z + Z^\dagger}{2i}.$$

Then

$$X^\dagger = \frac{Z^\dagger - Z}{2} = -X,$$

and

$$Y^\dagger = \left(\frac{Z + Z^\dagger}{2i}\right)^\dagger = \frac{Z^\dagger + Z}{-2i} = -Y.$$

So both X and Y are skew-Hermitian. Since $\text{tr } Z = 0$, also $\text{tr } Z^\dagger = \overline{\text{tr } Z} = 0$, and hence

$$\text{tr } X = 0, \quad \text{tr } Y = 0.$$

Thus $X, Y \in \mathfrak{su}(n)$. Finally,

$$X + iY = \frac{Z - Z^\dagger}{2} + i\frac{Z + Z^\dagger}{2i} = \frac{Z - Z^\dagger}{2} + \frac{Z + Z^\dagger}{2} = Z.$$

Therefore every element of $\mathfrak{sl}(n, \mathbb{C})$ lies in $\mathfrak{su}(n) + i\mathfrak{su}(n)$.

This proves, as complex Lie algebras,

$$\boxed{\mathfrak{sl}(n, \mathbb{C}) \cong \mathfrak{su}(n) \otimes_{\mathbb{R}} \mathbb{C}.}$$

Part (b): angular momentum and $\mathfrak{sl}(2, \mathbb{C})$

On $L^2(\mathbb{R}^3)$, the position and momentum operators satisfy the canonical commutation relations

$$[x_i, x_j] = 0, \quad [p_i, p_j] = 0, \quad [x_i, p_j] = i\delta_{ij}.$$

Here

$$p_j = -i\frac{\partial}{\partial x_j}.$$

The angular momentum operators are

$$L_i = \sum_{j,k} \epsilon_{ijk} x_j p_k.$$

Equivalently,

$$\mathbf{L} = \mathbf{x} \times \mathbf{p}.$$

Explicitly,

$$L_1 = x_2 p_3 - x_3 p_2,$$

$$L_2 = x_3 p_1 - x_1 p_3,$$

$$L_3 = x_1 p_2 - x_2 p_1.$$

Using the canonical commutation relations, one obtains the standard angular momentum commutation relations

$$[L_i, L_j] = i \sum_k \epsilon_{ijk} L_k.$$

Let us briefly indicate the structure of the computation. Since the L_i are quadratic expressions in the x_j and p_k , the commutator is computed using

$$[AB, C] = A[B, C] + [A, C]B$$

and the basic relations above. All terms reduce to the commutators $[x_i, p_j] = i\delta_{ij}$, and the result is precisely the formula above.

Now define

$$L_+ = L_1 + iL_2, \quad L_- = L_1 - iL_2, \quad L_z = L_3.$$

Then the commutation relations become

$$[L_z, L_+] = L_+, \quad [L_z, L_-] = -L_-, \quad [L_+, L_-] = 2L_z.$$

These are exactly the standard $\mathfrak{sl}(2, \mathbb{C})$ commutation relations if we identify the standard generators

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

with

$$h \mapsto 2L_z, \quad e \mapsto L_+, \quad f \mapsto L_-.$$

Indeed,

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h,$$

and under the assignment above these become

$$[2L_z, L_+] = 2L_+,$$

$$[2L_z, L_-] = -2L_-,$$

$$[L_+, L_-] = 2L_z.$$

Thus the angular momentum operators define a representation of $\mathfrak{sl}(2, \mathbb{C})$ on a suitable dense domain in $L^2(\mathbb{R}^3)$, for example on smooth compactly supported functions away from the usual analytic subtleties. Formally, we write

$$\boxed{\mathfrak{sl}(2, \mathbb{C}) \longrightarrow \text{End}(L^2(\mathbb{R}^3)), \quad h \mapsto 2L_z, \quad e \mapsto L_+, \quad f \mapsto L_-}.$$

Remark. Some physics conventions include a factor of $\hbar$ in the momentum operator, $p_j = -i\hbar\partial_j$. In that convention the commutation relations contain factors of $\hbar$, and one rescales the generators accordingly. In this exercise the definition $p_j = -i\partial_j$ corresponds to setting $\hbar = 1$.

Part (c): decomposition using spherical harmonics

We now describe the representation-theoretic decomposition of $L^2(\mathbb{R}^3)$. Using spherical coordinates,

$$x = (r, \theta, \varphi),$$

the measure on $\mathbb{R}^3$ is

$$d^3x = r^2 dr d\Omega,$$

where $d\Omega$ is the standard measure on the sphere S^2 . Therefore there is a Hilbert space decomposition

$$L^2(\mathbb{R}^3) \cong L^2(\mathbb{R}_{>0}, r^2 dr) \otimes L^2(S^2).$$

The rotation group $SO(3)$, and hence its Lie algebra, acts only on the angular variable. It does not change the radial coordinate r .

The spherical harmonics satisfy

$$L^2(S^2) = \widehat{\bigoplus_{\ell=0}^{\infty} \mathcal{H}_{\ell}},$$

where

$$\mathcal{H}_{\ell} = \text{span}\{Y_{\ell}^m \mid m = -\ell, -\ell + 1, \dots, \ell\}.$$

The dimension of $\mathcal{H}_{\ell}$ is

$$\dim \mathcal{H}_{\ell} = 2\ell + 1.$$

The angular momentum operators act on spherical harmonics by

$$L_z Y_{\ell}^m = m Y_{\ell}^m,$$

$$L_+ Y_{\ell}^m = \sqrt{\ell(\ell+1) - m(m+1)} Y_{\ell}^{m+1},$$

and

$$L_- Y_{\ell}^m = \sqrt{\ell(\ell+1) - m(m-1)} Y_{\ell}^{m-1}.$$

Here we are using the convention $\hbar = 1$. These formulas show that, for a fixed ℓ , the operators L_z, L_+, L_- move inside the finite-dimensional space $\mathcal{H}_{\ell}$. They do not change ℓ . Hence each $\mathcal{H}_{\ell}$ is a subrepresentation.

Moreover, $\mathcal{H}_{\ell}$ is irreducible. Indeed, Y_{ℓ}^{ℓ} is a highest weight vector: it is killed by L_+ , and repeated applications of L_- produce all the vectors

$$Y_{\ell}^{\ell}, Y_{\ell}^{\ell-1}, \dots, Y_{\ell}^{-\ell}.$$

Thus any nonzero invariant subspace containing one weight vector must contain all of $\mathcal{H}_{\ell}$. This is the standard irreducible $(2\ell + 1)$ -dimensional representation of $\mathfrak{sl}(2, \mathbb{C})$, with highest weight 2ℓ in the usual $\mathfrak{sl}(2)$ normalization.

Therefore

$$L^2(\mathbb{R}^3) \cong L^2(\mathbb{R}_{>0}, r^2 dr) \otimes \widehat{\bigoplus_{\ell=0}^{\infty} \mathcal{H}_{\ell}}.$$

Equivalently,

$$L^2(\mathbb{R}^3) \cong \widehat{\bigoplus_{\ell=0}^{\infty} (L^2(\mathbb{R}_{>0}, r^2 dr) \otimes \mathcal{H}_{\ell})}.$$

In words: the angular part decomposes into irreducible angular momentum representations $\mathcal{H}_{\ell}$, while the radial part is a multiplicity space. The Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ acts nontrivially on $\mathcal{H}_{\ell}$ and trivially on the radial factor.

This is the representation-theoretic content behind the familiar separation of variables in central potentials: the quantum number ℓ labels irreducible representations of the rotation algebra, and m labels a basis inside the $(2\ell + 1)$ -dimensional representation.